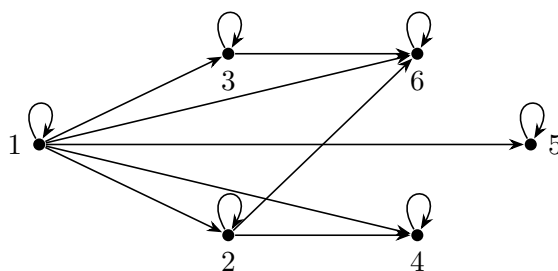


11 Relations

11.0 Relations

2. Let $A = \{1, 2, 3, 4, 5, 6\}$. Write out the relation R that expresses $\mid$ (divides) on A . Then illustrate it with a diagram.

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$



6. Congruence modulo 5 is a relation on the set $A = \mathbb{Z}$. In this relation xRy means $x \equiv y \pmod{5}$. Write out the set R in set-builder notation.

$$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : 5 \mid (x - y)\}.$$

8. Let $A = \{1, 2, 3, 4, 5, 6\}$. Observe that $\emptyset \subseteq A \times A$, so $R = \emptyset$ is a relation on A . Draw a diagram for this relation.

Since $R = \emptyset$, no pair (x, y) belongs to R , so the diagram has a node for each element of A and no arrows at all.



10. Consider the subset $R = (\mathbb{R} \times \mathbb{R}) - \{(x, x) : x \in \mathbb{R}\} \subseteq \mathbb{R} \times \mathbb{R}$. What familiar relation on $\mathbb{R}$ is this? Explain.

The relation is *not equal* ($\neq$). The set contains every pair (x, y) of real numbers. Removing the pair (x, x) leaves exactly the pairs where x and y differ.

11.1 Properties of Relations

2. Consider the relation $R = \{(a, b), (a, c), (c, c), (b, b), (c, b), (b, c)\}$ on the set $A = \{a, b, c\}$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why.

This is not reflexive because $(a, a) \notin R$.

This is not symmetric because $(a, b) \in R$ but $(b, a) \notin R$.

This is transitive: checking every chain xRy, yRz of pairs in R confirms that $(x, z) \in R$ in each case.

4. Let $A = \{a, b, c, d\}$. Suppose R is the relation

$$R = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (a, c), (c, a), (a, d), (d, a), (b, c), (c, b), (b, d), (d, b), (c, d), (d, c)\}.$$

Is R reflexive? Symmetric? Transitive? If a property does not hold, say why.

This is reflexive because xRx holds for every $x \in A$.

This is symmetric because xRy and yRx both hold for every $x, y \in A$.

This is transitive because xRz holds for every $x, z \in A$, regardless of y .

6. Consider the relation $R = \{(x, x) : x \in \mathbb{Z}\}$ on $\mathbb{Z}$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why. What familiar relation is this?

Here xRy if and only if $x = y$, so R is the equality relation $=$ on $\mathbb{Z}$.

This is reflexive because $x = x$ for every $x \in \mathbb{Z}$.

This is symmetric because $x = y$ implies $y = x$.

This is transitive because $x = y$ and $y = z$ imply $x = z$.

8. Define a relation on $\mathbb{Z}$ as xRy if $|x - y| < 1$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why. What familiar relation is this?

Since $x - y$ is an integer and the only integer with absolute value less than 1 is 0, we have xRy if and only if $x - y = 0$, that is, $x = y$. So R is the equality relation $=$ on $\mathbb{Z}$.

This is reflexive because $x = x$ for every $x \in \mathbb{Z}$.

This is symmetric because $x = y$ implies $y = x$.

This is transitive because $x = y$ and $y = z$ imply $x = z$.

10. Suppose $A \neq \emptyset$. Since $\emptyset \subseteq A \times A$, the set $R = \emptyset$ is a relation on A . Is R reflexive? Symmetric? Transitive? If a property does not hold, say why.

This is not reflexive. Pick any $x \in A$, then $(x, x) \notin R$ because $R = \emptyset$.

This is symmetric because it is impossible to find a $(x, y) \in R$ for which $(y, x) \notin R$.

This is transitive because $xRy \wedge yRz$ implies xRz for every $x, y, z \in A$.

12. Prove that the relation $|$ (divides) on the set $\mathbb{Z}$ is reflexive and transitive. (Use Example 11.8 as a guide if you are unsure of how to proceed.)

Proof. First we will show that $|$ (divides) is reflexive. Take any integer $x \in \mathbb{Z}$. Since $x = x \cdot 1$ and $1 \in \mathbb{Z}$, the definition of divisibility gives $x | x$. Thus $|$ is reflexive.

Next, we will show that $|$ is transitive. For this, we must show that if $x | y$ and $y | z$ then $x | z$. Suppose $x | y$ and $y | z$. This means $y = xm$ and $z = yn$ for some integers m and n . Substituting xm for y we get $z = (xm)n = x(mn)$, and by definition of divisibility, we see $x | z$.

The past two paragraphs show that $|$ is reflexive and transitive on $\mathbb{Z}$. □

14. Suppose R is a symmetric and transitive relation on a set A , and there is an element $a \in A$ for which aRx for every $x \in A$. Prove that R is reflexive.

16. Define a relation R on $\mathbb{Z}$ by declaring that xRy if and only if $x^2 \equiv y^2 \pmod{4}$. Prove that R is reflexive, symmetric and transitive.

18. The table on page 179 shows that relations on $\mathbb{Z}$ may obey various combinations of the reflexive, symmetric and transitive properties. In all, there are $2^3 = 8$ possible combinations, and the table shows 5 of them. (There is some redundancy, as $\leq$ and $|$ have the same type.) Complete the table by finding examples of relations on $\mathbb{Z}$ for the three missing combinations.

11.2 Equivalence Relations

2. Let $A = \{a, b, c, d, e\}$. Suppose R is an equivalence relation on A . Suppose R has two equivalence classes. Also aRd , bRc and eRd . Write out R as a set.
4. Let $A = \{a, b, c, d, e\}$. Suppose R is an equivalence relation on A . Suppose also that aRd and bRc , eRa and cRe . How many equivalence classes does R have?
6. There are five different equivalence relations on the set $A = \{a, b, c\}$. Describe them all. Diagrams will suffice.
8. Define a relation R on $\mathbb{Z}$ as xRy if and only if $x^2 + y^2$ is even. Prove R is an equivalence relation. Describe its equivalence classes.
10. Suppose R and S are two equivalence relations on a set A . Prove that $R \cap S$ is also an equivalence relation. (For an example of this, look at Figure 11.2. Observe that for the equivalence relations R_2, R_3 and R_4 , we have $R_2 \cap R_3 = R_4$.)
12. Prove or disprove: If R and S are two equivalence relations on a set A , then $R \cup S$ is also an equivalence relation on A .
14. Suppose R is a reflexive and symmetric relation on a finite set A . Define a relation S on A by declaring xSy if and only if for some $n \in \mathbb{N}$ there are elements $x_1, x_2, \dots, x_n \in A$ satisfying xRx_1 , x_1Rx_2 , x_2Rx_3 , $x_3Rx_4, \dots, x_{n-1}Rx_n$, and x_nRy . Show that S is an equivalence relation and $R \subseteq S$. Prove that S is the unique smallest equivalence relation on A containing R .

11.3 Equivalence Classes and Partitions

2. List all the partitions of the set $A = \{a, b, c\}$. Compare your answer to the answer to Exercise 6 of Section 11.2.
4. Suppose P is a partition of a set A . Define a relation R on A by declaring xRy if and only if $x, y \in X$ for some $X \in P$. Prove R is an equivalence relation on A . Then prove that P is the set of equivalence classes of R .

11.4 The Integers Modulo n

2. Write the addition and multiplication tables for $\mathbb{Z}_3$.
4. Write the addition and multiplication tables for $\mathbb{Z}_6$.
6. Suppose $[a], [b] \in \mathbb{Z}_6$ and $[a] \cdot [b] = [0]$. Is it necessarily true that either $[a] = [0]$ or $[b] = [0]$?
8. Suppose $[a], [b] \in \mathbb{Z}_n$, and $[a] = [a']$ and $[b] = [b']$. Alice adds $[a]$ and $[b]$ as $[a] + [b] = [a + b]$. Bob adds them as $[a'] + [b'] = [a' + b']$. Show that their answers $[a + b]$ and $[a' + b']$ are the same.